



KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

SPH 101: ELECTRICITY AND MAGNETISM I

2ND SEMESTER 2023/2024

TIME: 2 HOURS

INSTRUCTIONS:

Answer Question ONE and any other TWO questions. Question one carries 30 marks while each of the others carry 20 marks.

Some constants which you may find useful

Magnitude of the charge of an electron, e

$$1.6 \times 10^{-19} \text{ C}$$

Acceleration due to gravity, g

$$10 \text{ ms}^{-2}$$

Value of $k = \frac{1}{4\pi\epsilon_0}$

$$9 \times 10^9 \text{ mF}^{-1}$$

Mass of an electron, m_e

$$9.11 \times 10^{-31} \text{ kg}$$

i. (a) (i) State Coulomb's law.

(2 marks)

(ii) What will be the line of action of the force on a charge placed on the perpendicular bisector of the line joining two equal and like charges? Explain your answer.

(3 marks)

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(b) An electron is moving in a circular orbit of radius 0.1m in a plane at right angles to a uniform magnetic field of $2.0 \times 10^{-5}\text{T}$. What is the speed of the electron?

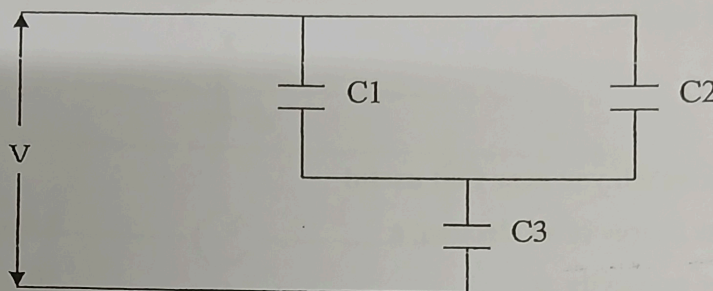
(5 marks)

(c) Find the magnitude and direction of the electric field which can balance the weight of an electron.

(5 marks)

(d)(i) In the following combination find the equivalent capacitance assuming that $C_1 = 2\mu\text{F}$, $C_2 = 3\mu\text{F}$, $C_3 = 4\mu\text{F}$ and $V = 30\text{ Volts}$.

(2 marks)



(ii) Find the magnitude of the charge on the plates of each capacitor. (3 marks)

(e) When a battery of emf 50V is connected to a load resistance of $10\ \Omega$, the terminal voltage is measured to be 45V . What is the value of the internal resistance of the battery?

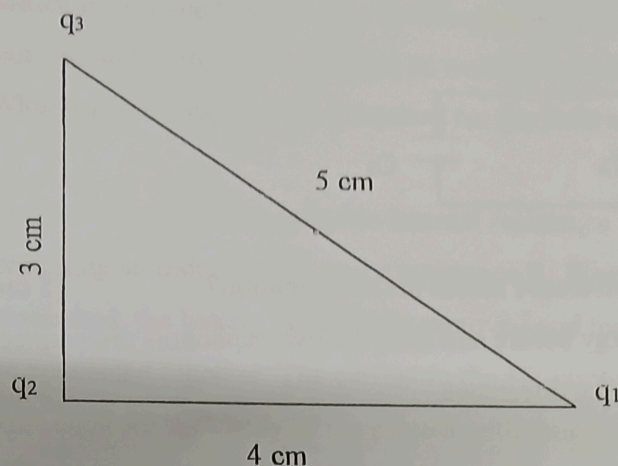
(5 marks)

(f) The material of an electric wire (length 1.0 m , cross sectional area $= 3.0 \times 10^{-6}\text{m}^2$) has a resistivity of $\rho_0 = 6.8 \times 10^{-5}\ \Omega\cdot\text{m}$ at $T_0 = 320\ ^\circ\text{C}$ and a temperature coefficient of resistivity $\alpha = 2.0 \times 10^{-3}/^\circ\text{C}$. Determine the resistance of the wire at a temperature of $360\ ^\circ\text{C}$.

(5 marks)

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2. (a) Three charges $q_1 = +4 \text{ nC}$, $q_2 = +5 \text{ nC}$ and $q_3 = -5 \text{ nC}$, are located as shown in the figure below.



Find the magnitude and direction of the net force on q_1 . (8 marks)

- (b) A ring-shaped conductor with radius a carries a total charge q , uniformly distributed around it. Find the electric field at a point P that lies on the axis of the ring at a distance x from the centre of the ring. (6 marks)

- (c) An electric dipole consists of two point charges, $+8 \text{ nC}$ and -8 nC placed 6 cm apart. Compute the potentials at

- (i) a point on the perpendicular bisector of the line joining the two charges,

(3 marks)

- (ii) a point along the axis of the dipole and 4 cm from the centre of the dipole towards the positive charge. (3 marks)

3. (a) Draw a circuit diagram for the potentiometer and explain its working.

(10 marks)

- (b) A galvanometer gives full scale deflection for a current of 0.03 A and has a resistance of 60Ω .

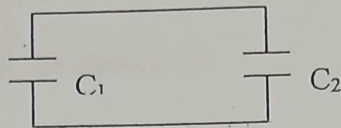
- (i) How can it be converted into a 5 A ammeter?

(5 marks)

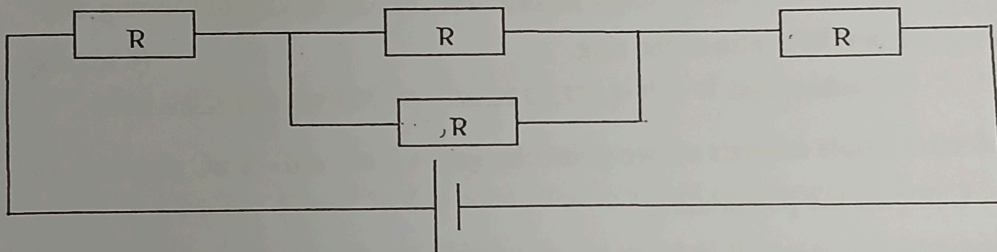
- (ii) How can it be converted into a 6 V voltmeter?

(5 marks)

4. (a) A capacitor $C_1 = 3\mu\text{F}$ is charged to a potential difference of 100 V. Charging battery is then removed and the capacitor is connected, as shown in the figure, to an uncharged capacitor $C_2 = 6\mu\text{F}$.



- (i) What is the potential difference across the combination? (5 marks)
- (ii) What is the stored energy before and after the two capacitors are connected together? Comment on the difference. (5 marks)
- (b) Apply Kirchhoff's laws to find the equivalent resistance of the network shown below. Assume that the resistance of each resistor R is 5Ω and the battery has emf $\mathcal{E}$. (5 marks)



- (c) A 200 W electric heater is designed to operate from 120 V lines.
- (i) What is its resistance? (2 marks)
- (ii) What current does it draw? (2 mark)
- (iii) If the line voltage drops to 100V, what power does the heater take? (1 mark)

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5.(a)(i) State Faraday's law of electromagnetic induction. (2 marks)

(ii) A 15 turn coil 12 cm in diameter has its axis parallel to a magnetic field of 0.5 T which is produced by a nearby electromagnet. The current in the electromagnet is cut off, and as the field collapses an average emf of 7V is induced in the coil.

What was the length of time required for the field to disappear? (4 marks)

(b) A battery of emf ξ and negligible internal resistance is switched on in a circuit containing an inductance L and a resistance R . When the equilibrium current is established, the battery is suddenly removed and the inductance and the resistance are connected together. The current in the LR circuit starts decaying. Derive an expression for the decay of the current with time in the circuit. Hence define the time constant and explain its physical significance. (10 marks)

(c) A series circuit containing a 6.0 H inductance and a 12.0 Ω resistance is connected to a 15 V battery of negligible internal resistance. When the equilibrium current is established, the battery is suddenly removed and the inductance and the resistance are connected together. The current in the LR circuit starts decaying. Find

(i) The initial rate of decay of the current in the circuit (2 mark)

(ii) The current 1 second after the decay started. (2 marks)

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SPH 191: ELECTRICITY AND MAGNETISM I

2ND SEMESTER 2022/2023

TIME: 2 HOURS

INSTRUCTIONS: Answer question **ONE** and any other **TWO** questions. Question one carries 24 marks and other questions 18 marks each.

The following information may be useful

$$\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ Nm}^2\text{C}^{-2}$$

1electron-volt (1 eV)	1.6×10^{-19} joules
Acceleration due to gravity, g	9.8 ms^{-2}
Electronic charge, e	$1.6 \times 10^{-19} \text{ C}$
Gravitational constant G	$6.7 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$
Mass of an electron, m_e	$9.1 \times 10^{-31} \text{ kg}$
Mass of a proton, m_p	$1.7 \times 10^{-27} \text{ kg}$
Mass of the earth	$5.2 \times 10^{24} \text{ kg}$
Permeability of free space, μ_0	$1.3 \times 10^{-6} \text{ Hm}^{-1}$
Permittivity constant, ϵ_0	$8.9 \times 10^{-12} \text{ Fm}^{-1}$
Radius of the earth	$6.4 \times 10^6 \text{ m}$
Resistivity of copper	$1.7 \times 10^{-8} \Omega\text{m}$
Speed of light	$3.0 \times 10^8 \text{ m/s}$

QUESTION ONE CONSISTS OF TWO SECTIONS ("A" AND "B"). YOU MUST ANSWER QUESTIONS FROM BOTH SECTIONS

SECTION A: SELECT AND WRITE THE LETTER CORRESPONDING TO THE MOST CORRECT ANSWER

1. a) An electrical insulator is a material:
- A. containing no electrons
 - B. through which electrons do not flow easily
 - C. that has more electrons than protons on its surface
 - D. cannot be a pure chemical element
 - E. must be a crystal

- b) Choose the correct statement concerning electric field lines:
- A. field lines may cross
 - B. field lines are close together where the field is large
 - C. field lines point away from a negatively charged particle
 - D. a charged point particle released from rest moves along a field line
 - E. none of these are correct
- c) The potential difference between two points is 100V. If a particle with a charge of 2C is transported from one of these points to the other, the magnitude of the work done is:
- A. 200 J B. 100 J C. 50 J D. 100 J E. 2 J
- d) An electron volt is:
- A. the force acting on an electron in a field of 1N/C
 - B. the force required to move an electron 1 meter
 - C. the energy gained by an electron in moving through a potential difference of 1 volt
 - D. the energy needed to move an electron through 1 meter in any electric field
 - E. the work done when 1 coulomb of charge is moved through a potential difference of 1 volt.
- e) Pulling the plates of an isolated charged capacitor apart:
- A. increases the capacitance
 - B. increases the potential difference
 - C. does not affect the potential difference
 - D. decreases the potential difference
 - E. does not affect the capacitance
- f) Two parallel-plate capacitors with different capacitance but the same plate separation are connected in series to a battery. Both capacitors are filled with air. The quantity that is the same for both capacitors when they are fully charged is:
- A. potential difference
 - B. stored energy
 - C. energy density
 - D. electric field between the plates
 - E. charge on the positive plate
- g) Current is a measure of:
- A. force that moves a charge past a point
 - B. resistance to the movement of a charge past a point
 - C. energy used to move a charge past a point
 - D. amount of charge that moves past a point per unit time
 - E. speed with which a charge moves past a point
- h) The current is zero in a conductor when no potential difference is applied because:
- A. the electrons are not moving
 - B. the electrons are not moving fast enough
 - C. for every electron with a given velocity there is another with a velocity of equal magnitude and opposite direction.
 - D. equal numbers of electrons and protons are moving together
 - E. otherwise Ohm's law would not be valid
- i) Of the following, the copper conductor that has the least resistance is:
- A. thin, long and hot
 - B. thick, short and cool
 - C. thick, long and hot
 - D. thin, short and cool
 - E. thin, short and hot

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j). Conductivity is:

- A. the same as resistivity, it is just more convenient to use for good conductors
- B. expressed in Ω^{-1}
- C. equal to $1/\text{resistance}$
- D. expressed in $(\Omega \text{ m})^{-1}$
- E. not a meaningful quantity for an insulator

(1 × 10 = 10 Marks)

SECTION B: SHOW ALL YOUR WORK ON THE PROVIDED ANSWER SHEET. Credit will be awarded for clear explanations and illustrations

1k) State the four facts, which are summarized in Coulomb's law.

(4 Marks)

1l) Define the term *time constant* for capacitors.

(2 Marks)

1m) Two equally charged particles are held $3.2 \times 10^{-3} \text{ m}$ apart and then released from rest. The initial acceleration of the first particle is observed to be 7.0 m/s^2 and that of the second to be 9.0 m/s^2 . If the mass of the first particle is $6.3 \times 10^{-7} \text{ kg}$, what is the mass of the second particle.

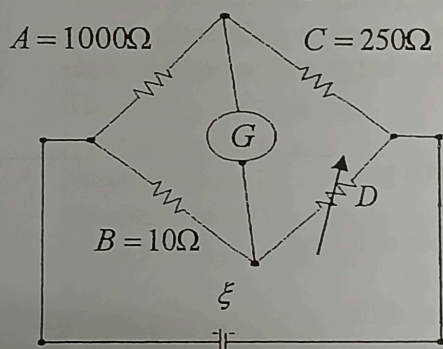
(4 Marks)

1n) Two particles are fixed to an x axis: particle 1 of charge $-2.00 \times 10^{-7} \text{ C}$ at $x = 6.00 \text{ cm}$ and particle 2 of charge $+2.00 \times 10^{-7} \text{ C}$ at $x = 21.0 \text{ cm}$. Midway between the particles, what is their net electric field in unit-vector notation?

(4 Marks)

2. a) Explain, giving examples, the phenomenon of 'electrical shock'.

(6 Marks)



b) Figure 1 shows a circuit diagram of a Wheatstone bridge. What is the value of the unknown resistor D given that the galvanometer reads zero deflection?

(2 Marks)

c) i) State Faraday's law of electromagnetic induction and write its mathematical form.

(2 Marks)

ii) State the factors on which the induced emf depends.

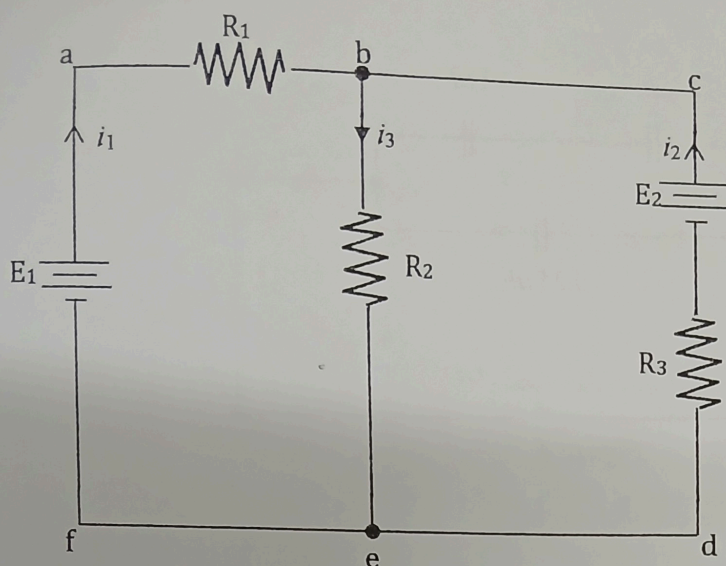
(2 Marks)

d) A moving coil galvanometer has a resistance r_g . Full-scale deflection (f.s.d) is produced by a current i_g . Explain, with the aid of a circuit diagram, how this galvanometer can be converted into an ammeter able to measure currents up to i_m , where i_m is greater than i_g . (6 Marks)

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3. a) i) State Kirchhoff's current and voltage laws.

(4 Marks)



ii) Find the currents through each branch of the network in figure 6, given that $R_1 = 27\Omega$, $R_2 = 33\Omega$, $R_3 = 66\Omega$, $E_1 = 9.0\text{V}$ and $E_2 = 6.0\text{V}$.

(8 Marks)

Figure 2

b) For a circuit containing an *emf* source E , a resistor of resistance R and a capacitor of capacitance C , show that the instantaneous charge Q on the capacitor when discharging after t seconds is given by the relationship, $Q = Q_0 e^{-t/RC}$.

(6 Marks)

4. a) The specifications in an LR circuit are as follows: inductance $L = 2.0\text{ H}$, the resistance $R = 20\ \Omega$ and the power supply $E = 100\text{V}$. If the internal resistance of the supply is $5\ \text{ohms}$, calculate the steady current flowing in the circuit.

(4 Marks)

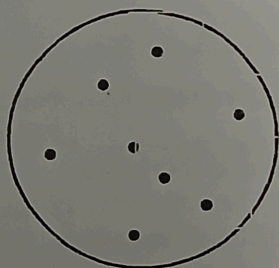


Figure 3

b) The lines of a time-varying magnetic flux density, B , are directed perpendicular to a circular loop as shown in figure 3. The radius of the circular loop is $10 \times 10^{-4}\text{ m}$. The initial value of the magnetic density, B_0 , is $4 \times 10^{-2}\text{ T}$, and B decreases at the rate of $2 \times 10^{-4}\text{ Ts}^{-1}$

i) What is the initial magnetic flux (Φ_0) through the loop?

(5 Marks)

ii) What is the induced e.m.f in the circular loop?

(5 Marks)

c) Two large, parallel, conducting plates are 12 cm apart and have charges of equal magnitude and opposite sign on their facing surfaces. An electrostatic force of $3.9 \times 10^{-15}\text{ N}$ acts on an electron placed anywhere between the two plates.

i) Find the electric field at the position of the electron.

(2 Marks)

ii) What is the potential difference between the plates?

(2 Marks)

- d) Figure 4 shows a circuit diagram having four capacitors and an *emf* source. Calculate the effective capacitance of figure 4. (3 Marks)

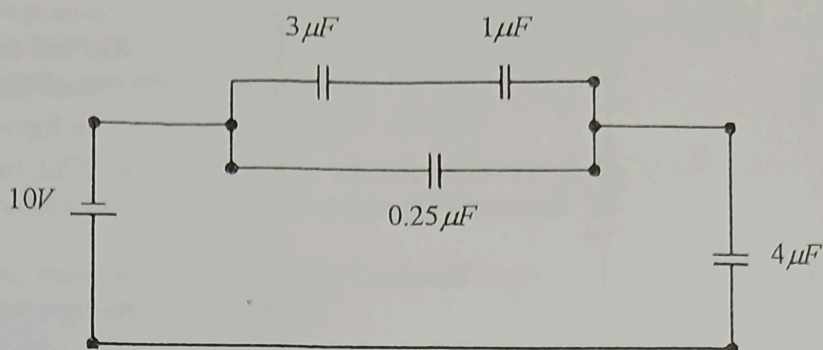


Figure 4

5. a) The diagram in Figure 3 is that of an electrical circuit.

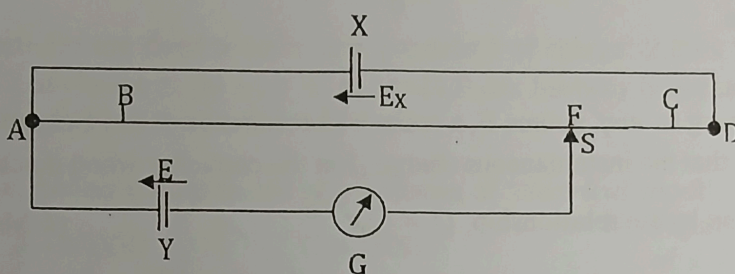


Figure 5

- Identify the circuit in figure 5 and state its main components. (3 Marks)
 - State two (2) applications of this circuit. (2 Marks)
 - Explain the measuring process of the circuit. (5 Marks)
- b) Calculate,
- The electric potential established by the nucleus of a hydrogen atom at the mean distance of the circulating electron ($r = 5.3 \times 10^{-11}\text{m}$). (2 Marks)
 - the electric potential energy of the atom when the electron is at this radius. (2 Marks)
- c) A conducting wire has a 1.0 mm diameter, a 2.0 m length, and a $50\text{ m}\Omega$ resistance. What is the resistivity of the material? (4 Marks)

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